

PMG 4101 Exam Notes

Agile & Waterfall - Complete Important Questions with Easy Answers

Software Methodology

Q. What is Software Methodology?

A step-by-step organized process for developing software. It defines what to do, when to do it, how to communicate, test and deliver software.

Q. Why is it important?

- Organized development
- Reduces errors
- Better communication
- Understands customer needs
- Saves time & cost
- Better quality
- Easier maintenance

Q. Name five methodologies.

Agile, Waterfall, Scrum, Spiral, RUP (also DevOps, Prototype, XP, RAD).

Agile Methodology

Q. What is Agile?

A flexible and iterative methodology. Software is delivered in short Sprints (1-4 weeks).

Q. Key Idea

Deliver Small → Deliver Fast → Get Feedback → Improve Continuously.

Q. Advantages

1. Minimal defects
2. Better communication
3. Handles changing requirements
4. Better software quality

Q. Disadvantages

1. Too many changes reduce focus
2. Less documentation
3. Time-consuming discussions
4. Needs experienced developers

Q. When to use Agile?

Changing requirements, customer feedback, complex projects, dynamic market, cross-functional teams.

Q. When NOT to use Agile?

Fixed requirements, heavy documentation, unavailable stakeholders, inexperienced team, fixed budget/scope/deadline.

Q. What is Sprint?

A Sprint is a short development cycle (1-4 weeks) that produces working software.

Waterfall Model

Q. What is Waterfall?

A linear model where each phase is completed before the next starts.

Q. Advantages

Easy to understand, strong documentation, clear requirements, less communication confusion.

Q. Disadvantages

No early customer feedback, testing at the end, difficult to change requirements, documentation-heavy.

Q. When to choose Waterfall?

Clear requirements, fixed scope, known technology, regulated industries, sequential development.

Q. When NOT to choose Waterfall?

Changing requirements, rapid iterations, experimental technology, continuous delivery, highly flexible projects.

Agile vs Waterfall

Agile	Waterfall
Flexible	Fixed
Iterative	Linear
Continuous Testing	Testing at End
Frequent Delivery	Final Delivery
Easy to Change	Difficult to Change
Customer Feedback	Limited Feedback

Hybrid Model

Q. What is Hybrid?

Hybrid combines Waterfall (Requirements & Architecture) with Agile (UI, Features & Enhancements).

Q. Banking Example

Requirements & documentation follow Waterfall; front-end and feature improvements follow Agile.

Most Expected Exam Questions

- 1. Define Agile.
- 2. Advantages & disadvantages of Agile.
- 3. When to use Agile?
- 4. When NOT to use Agile?
- 5. Define Waterfall.
- 6. Advantages & disadvantages of Waterfall.
- 7. When to choose Waterfall?
- 8. Agile vs Waterfall.
- 9. Choose a methodology for a given scenario with justification.
- 10. Explain Hybrid methodology with example.